THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY

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Bondi showed in 1957 that bodies of opposite *active* gravitational mass self-accelerate: the negative chases the positive it repels, and the pair runs off together. We evolve this “Bondi dipole” in 3+1 numerical relativity: two complex scalars share identical Klein–Gordon dynamics; only the phantom enters Einstein’s equations with a minus sign—inertial and passive masses positive, active mass negative. Across twenty-seven evolutions, **a mass-matched pair released at rest accelerates as a unit, at constant rate, with zero total momentum and constant separation**: the midpoint moves 3.00 ± 0.01 by $t = 200$, the signed momentum cancels to $\lesssim 1\%$, and the pair reaches $0.056c$ by $t = 400$ with the rate steady to 2%. It is gravity: $a \propto d^{-2.03\pm0.01}$ over $d = 8$–20, $a\,d^2$ returns the stars’ mass, and the pull follows the partner’s mass. Swapping the sectors inverts the acceleration to two parts in 10^5; every control is null, same-sign pairs holding their centroids to $\lesssim 8\times10^{-4}$ while merging. The phantom star is, to our knowledge, **the first stable asymptotically flat body of negative ADM mass evolved in numerical relativity**.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive and active gravitational mass and noting the consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body’s contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Forward [5] showed “in tedious detail” that the self-accelerating pair violates neither momentum nor energy conservation, for equal and unequal mass magnitudes alike, so objections to negative mass must be sought elsewhere [6, 7].

Why put this in full numerical relativity? Because nothing in Einstein’s equations forces mass to be positive—positivity is a theorem only once energy conditions are imposed [27, 28]—and modern physics keeps producing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, phantom dark energy [8]. Every traversable wormhole [9] and warp spacetime [10] needs matter of exactly this kind, and the standing objection to all of them is Bondi’s runaway itself—matter that accelerates for free looks like a perpetual-motion machine [6, 7]. That objection has only ever been examined for point particles and prescribed metrics; the nonlinear evolutions that exist for energy-condition-violating matter—ghost-scalar wormholes in spherical symmetry [15, 30], static phantom-field stars [31], and the 3+1 collapse of an Alcubierre warp bubble [11]—concern single objects. Whether the runaway survives full nonlinear gravity—finite bodies, backreaction, radiation—is a checkable question, and we are not aware of a previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

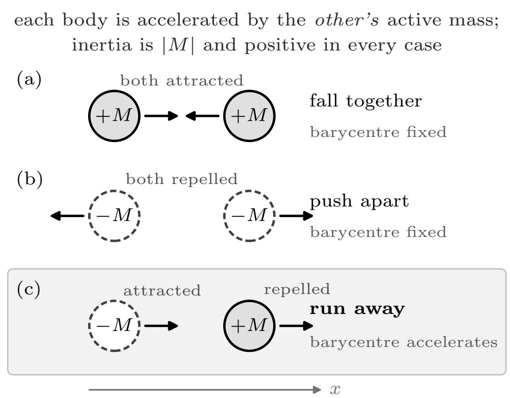

FIG. 1. Bondi’s sign rules for two bodies released at rest, in the style of Forward’s analysis [5]. Every body has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract and fall together; (b) two negative-mass stars repel; in both the forces oppose and the pair barycentre stays put. (c) The mixed pair—the Bondi dipole: both forces point the *same* way, so the pair accelerates while the system carries zero total momentum in the signed bookkeeping. Each row is a run family of Table I; in our field-theoretic realization the same-sign rows additionally feel their shared field’s own short-range attraction (Sec. V E), which the mixed row, by construction, cannot.

The question has acquired observational stakes: negative-mass binaries have been proposed as sources of anomalous gravitational-wave signatures—anti-chirps, dispersal, runaway motion—turning Bondi’s paradox into an exclusion channel for real detectors [39], while Nojiri and Odintsov [13] find that a bound positive–negative system *can* form for a compact mass immersed in a cosmological fluid with negative cosmological constant. Our

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asymptotically flat, isolated pair is the complementary case, and it does not bind—it runs away.

Figure 2 sets the programme of this paper against the point-mass argument it tests. We build a minimal field-theoretic realization of Bondi's sign structure (Sec. II); solve dressed soliton stars of both signs as constraint-satisfying initial data, with the two construction requirements that decide whether the measurement is possible at all (Sec. III); run a falsifiable matrix of twenty-seven evolutions (Sec. IV); report the measurement—a constant-rate, zero-momentum, constant-separation runaway obeying $a = \bar{M}/d^2$, with every control null (Sec. V); bound the gravitational-wave content on a doubled box (Sec. VI); and state the systematics with their measured sizes (Sec. VII). Every evolution ran on GPUs with `GRTeclyn` [24], the AMReX-based [26] GPU port of `GRChombo` [23]; with the wormhole campaign of Ref. [40] these are, to our knowledge, the first evolutions of energy-condition-violating matter in GPU-native numerical relativity, the warp-bubble collapse of Ref. [11] being the closest CPU-side precedent. Throughout, $G = c = 1$, the scalar mass sets the unit ($m = 1$), X is the pair midpoint (the mean of the two sector barycentres), $\Delta X = X(t) - X(0)$ its drift, and d the instantaneous separation. Runs are named for the sectors they contain, `P` for a canonical star and `M` for a phantom one: `PM` is the mixed pair, `PP` and `MM` the same-sign controls, `MP` the mixed pair with the sectors swapped in place.

II. THE BICOMPLEX SCALAR MODEL

The matter must keep canonical *dynamics*—positive inertial and passive mass—while only its gravitational *source* changes sign; it must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, since its own gravity pushes it apart. Each obvious candidate fails one of these: the negative-mass Schwarzschild spacetime is a naked singularity with exponentially growing smooth linear perturbations [14] and contains no matter from which to build initial data; a negative-energy fluid has no equilibrium to sit in, Bonnor's static spheres [4] demanding an inverted interior with no non-gravitational restoring force; and a ghost scalar reverses its field dynamics along with its source, flipping inertial and passive masses too (Forward's all-negative matter [5], not the signature under test), and such fields collapse or explode in evolution [15, 30, 40].

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We take two independent complex scalars, Φ_+ (*canonical*) and Φ_- (*phantom*), on one spacetime, sharing the standard sextic potential of Q-ball and solitonic boson-star studies [16–18],

$$V(|\Phi|^2) = \tfrac{1}{2}m^2|\Phi|^2 - \tfrac{1}{4}\lambda|\Phi|^4 + \tfrac{1}{6}\mu|\Phi|^6, \tag{1}$$

with $\lambda, \mu > 0$: the mass term makes the field oscillate, the attractive quartic lets it clump, the repulsive sextic stops the clumping short of collapse. Sharing one potential makes the matter physics of the two sectors *identical*, so the single sign introduced next is the only possible cause of any asymmetry the simulations show.

That sign lives in one place—the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi\left(T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]\right), \tag{2}$$

while the field equations of *both* sectors keep their canonical form,

$$\Box\,\Phi_\pm = 2\,\frac{\partial V}{\partial|\Phi|^2}\,\Phi_\pm, \tag{3}$$

the factor 2 fixed by the $\frac{1}{2}$-normalized kinetic term. In 3+1 language the sources seen by normal observers accumulate per sector with a sign $\epsilon_\pm = \pm 1$,

$$\rho = \sum_{s=\pm}\epsilon_s\,\rho[\Phi_s], \qquad S_i = \sum_{s=\pm}\epsilon_s\,S_i[\Phi_s], \tag{4}$$

and likewise for S_{ij}, each bracket being the standard scalar expression. The model derives from

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-]\right], \tag{5}$$

with $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_\mu\Phi^*\nabla^\mu\Phi - V(|\Phi|^2)$: a sector's overall Lagrangian sign cancels from its Euler–Lagrange equations—hence Eq. (3)—but survives the metric variation—hence Eq. (2). Bianchi consistency is automatic (each sector's stress tensor is separately conserved on shell, so the signed sum is too); well-posedness is inherited (the sector sign enters CCZ4 only through algebraic source terms); and the mass signature is Bondi's: the phantom *moves* canonically—positive inertial and passive mass—but *gravitates* with the minus sign of Eq. (2), so its active mass, and its ADM mass, are negative.

The model has exactly two interaction channels, and the run matrix is built to isolate them. The action carries no cross term between Φ_+ and Φ_-—each potential depends on its own $|\Phi_s|^2$ alone—so **two stars of different sectors interact only through the metric**: gravity is the mixed pair's only channel, by construction. Two stars of the *same* sector, by contrast, live in one complex field, whose self-interaction gives overlapping lumps a direct, short-range attraction on top of gravity. Section V E measures that field channel—it is strong, and blind to the sign of the mass—and it is precisely why the mixed pair, which cannot have it, is the clean Bondi configuration. Place one star of each kind side by side and both gravitational accelerations point from phantom toward canonical: Newton's third law "fails" in the precise sense that the two forces point the same way, and no conservation law breaks, because the conserved momentum belongs to the signed stress-energy of Eq. (4).

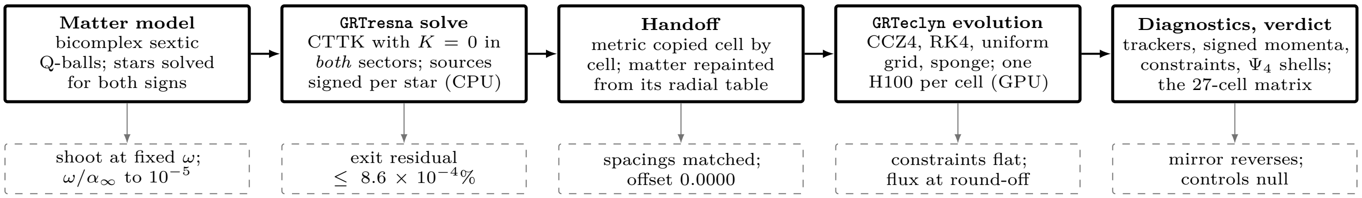

FIG. 2. What every cell of the campaign runs through (solid) and the check that gates each step (dashed); Bondi's argument stops before the first box, with no matter solved and no dynamics evolved. The two handoff requirements of Sec. III B are where the measurement is won or lost, and the last box is the falsifiable matrix of Table I.

The phantom sector violates the null energy condition by construction, and quantum viability is not claimed—a sector entering gravity with negative energy admits graviton-mediated vacuum decay [32, 33]—so the model is a classical probe of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8].

III. INITIAL DATA: DRESSED SOLITONS

A. The stars

A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and ω labels the solution family: $\omega = m$ is the free-field limit, $1 - \omega$ the binding energy per unit mass. The ratio λ^2/μ sets the lowest attainable frequency, $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$ [17]; all runs use $\lambda^2/\mu = 4.8$ ($\lambda = 10240$, $\mu \simeq 2.18 \times 10^7$, amplitude $\simeq$0.019), so $\omega_{\min} = 0.316$.

A flat-space soliton cannot simply be placed on an initial slice: the constraints are violated by a bare Q-ball, and once its own gravity acts the profile is no longer an eigenstate. The star we need is solved with its gravity *on*: the boson-star problem [19–21] with a sextic self-interaction, posed for both source signs. With $\Phi = \varphi_0(r)e^{-i\omega t}$ and a conformally flat, maximally sliced metric, Eqs. (2)–(3) reduce to coupled ODEs for $(\psi, \alpha, \varphi_0)$; **the phantom star is obtained by flipping the sign of the matter sources in the metric equations only**, the Klein–Gordon line keeping its canonical form. The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core, and negative ADM mass. We shoot at fixed frequency, bisecting the central amplitude and outer-iterating until ω/α_∞ matches the request to $\sim$$10^{-5}$.

Figure 3 shows both families. The campaign pair takes the canonical star at $\omega = 0.750$, $M_{\rm ADM} = +0.014350$, and the phantom at $\omega = 0.7603$, $M_{\rm ADM} = -0.014295$: mass-matched to 0.39%, because Bondi's point-mass limit is governed by each body's exterior field, hence its ADM mass, alone. The pair mean $\bar{M} \equiv \frac{1}{2}(M_+ + |M_-|) = 0.014322$ is what the midpoint acceleration measures. The canonical star has rms radius 4.34 and compactness $M/R \sim 3 \times 10^{-3}$—two orders below what any horizon needs—and a four-point survey ($\omega = 0.75$–0.90, $t = 120$) confirms the family static where it is used: the minimum lapse moves by at most 1.2×10^{-3} from birth. Each star is seeded from its own profile table with its own frequency, and the conjugate momentum carries the solved lapse, $\Pi = -(\omega/\alpha(r))\,\varphi_0(r)$, so the harmonic phase starts on its eigenstate.

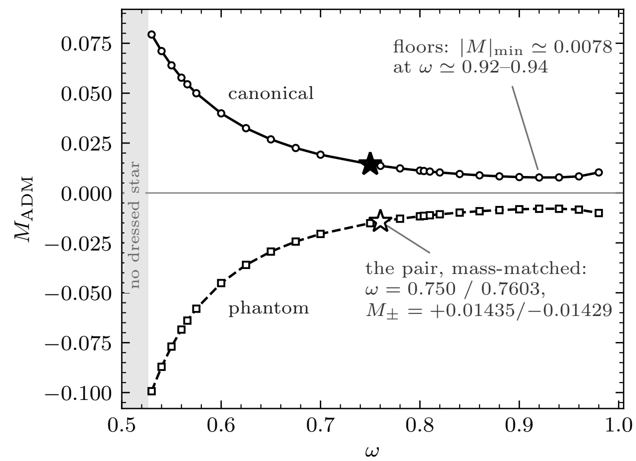

FIG. 3. ADM masses of the spherical stars against oscillation frequency at the working couplings. Lower ω means deeper binding and larger $|M_{\rm ADM}|$; at equal frequency the phantom star is the heavier in magnitude, its repulsive self-gravity spreading it over more volume. Stars: the campaign pair—canonical at $\omega = 0.750$, phantom retuned to $\omega = 0.7603$ so the mass magnitudes match to 0.4%. Both branches have a *floor*: no star on either side is lighter than $\simeq$0.54 of the campaign mass, and neither branch has a bound star at $\omega = 1$—the bound that shapes the mass-scaling ladder of Sec. V F.

B. Two construction requirements

`GRTresna` [25] then re-solves the constraints for the pair (CTTK family [35], sources summed with their signs star by star). Two choices in this step decide whether the runaway is measurable at all. Both were established by direct testing, and we state them because any reproduction will face them.

Both sectors must be built on the same slicing. The CTTK algebraic ansatz ties the initial K to the local energy density, $K \propto \sqrt{\rho}$—imaginary for $\rho < 0$, so a phantom sector forces the maximal-slicing ($K = 0$) path. Letting the canonical sector default to the algebraic path builds the two stars by *different methods*: the canonical star is then born inside a coherent contraction, $\max|K| \approx 0.1$ at $t = 0$—the ansatz formula evaluated at the core density, four orders of magnitude above the phantom's 10^{-5}—and is squeezed through its first two field periods. We therefore construct *both* sectors with $K = 0$, which for this matter also converges better; the birth kick vanishes ($\max|K| \to 10^{-5}$, landing on the phantom's value) and the two constructions differ in nothing but the sign of ρ.

The solve grid must share the evolution grid's cell centres. The elliptic solve runs on its own (wider) box; if its spacing differs from the evolution's, the solved metric lands on the evolution grid displaced by a fraction of a cell while the matter is repainted exactly—so every star is born off the centre of its own well, the canonical falling back down it and the phantom pushed up it, a directional artefact that does not converge away and was comparable to the signal. Matching the spacings makes the transfer a straight copy: the solved metric moves cell by cell (a piecewise-constant transfer, exact once the cell centres coincide) while the matter is repainted analytically from its radial profile table. The metric-minus-matter centroid offset is then 0.0000 on every cell here, and the lone-star drift below is 269× smaller than on mismatched grids. All solves run at the evolution spacing on a doubled box ($L = 128$), uniform, 32 MPI ranks; the mixed-pair solves exit at Hamiltonian error $\leq 8.6 \times 10^{-4}\%$, tightened with the grid to $8.7 \times 10^{-5}\%$ on the finest rung (Appendix A).

All pair runs seed the two lumps with $U(1)$ phases 0 and π. For the mixed pair this is inert (the sectors are independent fields, so their relative phase is pure gauge); in the same-sector controls both lumps live in *one* complex field, where superposed boson-star data carries known phase-dependent systematics that the constraint re-solve reduces but does not eliminate [34]—one reason the same-sign cells are used as momentum nulls, not as precision force measurements.

IV. THE CAMPAIGN

A. Numerics

Evolutions use `GRTeclyn`: CCZ4 [22] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [36], a Gamma-driver shift [37, 38] with $\eta = 1$, fourth-order stencils, RK4, Kreiss–Oliger dissipation $\sigma = 2$, Sommerfeld outer boundaries, and—because massive-field radiation reflects from massless-wave boundaries—an absorbing sponge layer over $r = 24$–32 (moved to 48–60 on the doubled box). The sponge works: the net inward flux through the accounting boundary stays at round-off ($\lesssim 10^{-7}$) even in the violent same-sign cells. The time step is $\Delta t = 0.02\,\Delta x$, an order of magnitude below the usual CCZ4 Courant factor and the dominant cost of the campaign (Appendix B); it is not conservatism but the sextic self-interaction's own stability limit—at $\Delta t = 0.2\,\Delta x$ the star disperses outright.

Every cell is *strictly uniform-grid* (`max_level` $= 0$), so the resolution ladder means exactly one thing—the cell size—and the aligned-grid handoff of Sec. III B is preserved verbatim. The main box is $L = 64$ at $N = 128$, 192, 256 ($\Delta x = 0.50$, 0.33, 0.25—about 9, 13 and 17 points across a stellar rms radius of 4.3); the wave-zone box doubles L at fixed $\Delta x = 0.25$ with extraction shells at $R = 16$–40. The reference separation is $d_0 = 10$, about twice the 90%-weight radius, and the separation scan spans $d_0 = 8$–20. Stars are released *at rest*—no boosts, no driving—so any drift is dynamical. All cells run to $t = 200$ (the long run to 400, the stability survey to 120), and every fitted acceleration a is twice the quadratic coefficient of $X(t)$ over $5 \leq t \leq 200$, the first five units excluded while the gauge settles; Sec. VII quantifies the (percent-level) sensitivity to that convention.

B. Diagnostics and the run matrix

Each sector is tracked two ways, continuously: a whole-domain barycentre with weight $w_s = (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, and a sub-cell core tracker at the sector's field peak. The headline quantity is the *pair midpoint* $X = \frac{1}{2}(\bar{x}^{(+)} + \bar{x}^{(-)})$: it is what Bondi's argument predicts to accelerate, it is defined identically for pairs and controls, and by symmetry it is exactly zero for every control—so the control rows of Table I measure the noise floor rather than assume one. The two trackers agree on the pair drift to 1%. Volume-integrated sector momenta, constraint norms, confinement, the $\ell = 2$ modes of Ψ_4, and energy-condition monitors are recorded alongside.

Table I is the experiment: each row is a falsifiable prediction of the sign structure, and a control drifting at the rate of the mixed pair would invalidate the physical origin of the runaway.

V. RESULTS

A. The runaway: constant rate, zero momentum, constant separation

Released at rest with the canonical star at $x = 37$ and the phantom at $x = 27$, **both stars accelerate toward $+x$ as a unit** (Figs. 4 and 5). On the finest grid the midpoint moves $\Delta X = +3.0016$ by $t = 200$ at fitted acceleration $a = 1.549 \times 10^{-4}$; a single parabola fits the whole trajectory. The $t = 400$ run (base grid, where $a = 1.45 \times 10^{-4}$) shows the rate is *sustained*: fitted over $[133, 266]$, $[300, 400]$ and $[350, 400]$ the acceleration is 1.451, 1.422 and 1.417×10^{-4}—steady to 2% across a factor three in elapsed time, with the pair at $0.056c$ and still gaining speed linearly when the run ends. There is no impulse, no saturation, and no secondary acceleration.

The two Bondi signatures frame that motion. **Displacement without momentum:** each sector's

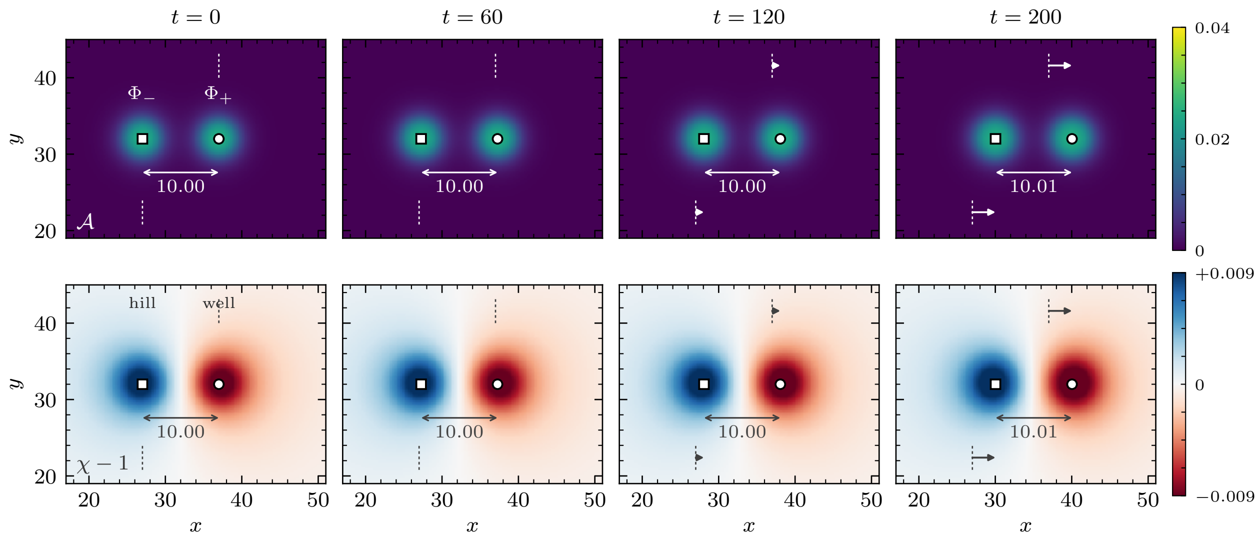

FIG. 4. The runaway in the headline cell (PM, $d_0 = 10$, $\Delta x = 0.25$), $z = 0$ plane, at $t = 0$, 60, 120, 200. Top: scalar-field activity $\mathcal{A} = \sum_s (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, both sectors, projected. Bottom: $\chi - 1$; the phantom star is a *hill* ($\chi > 1$, blue, left) and the canonical star a *well* ($\chi < 1$, red, right)—the model's one sign made visible. The colour bars carry the stills' locked scales, symmetric about $\chi - 1 = 0$ so that the sign divide is explicit. Open markers: the tracked cores (circle Φ_+, square Φ_-); dashed ticks mark the release positions $x = 37$ and 27, arrows the displacement since release, and the bracketed number is the measured gap. Both stars move toward $+x$ at constant acceleration while the gap holds to 1%.

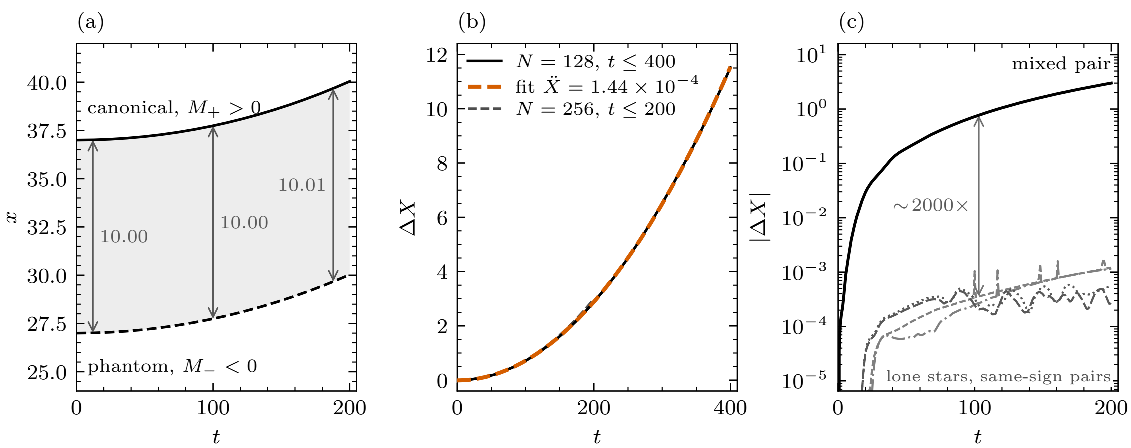

FIG. 5. (a) Core worldlines in the headline cell ($\Delta x = 0.25$): the phantom chases the canonical star it repels; the gap is marked at three times. (b) Midpoint drift of the $t = 400$ run (solid) with the single constant-acceleration fit over $5 \leq t \leq 400$ (orange dashed, as for every fitted curve in this paper)—one parabola holds for the whole run—and the finest-grid $t \leq 200$ cell (dashed). (c) The signal against every null on a log scale: the mixed pair (barycentre midpoint) against the lone stars (core tracker; the off-centre phantom is the sharper test) and the same-sign pair centroids—a separation of 3–4 orders of magnitude, sustained for the whole run.

volume-integrated momentum grows steadily—both reach 5.3×10^{-4} by $t = 200$, the scale set by $M|v|$—while the *signed* total of Eq. (4) cancels to 3.8×10^{-6}, under 1% of either term, and that residual *converges toward zero* with the grid (3.7×10^{-5} at $N = 128$, 3.8×10^{-6} at $N = 256$). The pair moves three units while carrying no net momentum, exactly as the signed bookkeeping demands. **Constant separation:** the barycentre gap ends at 9.915 and the core gap at 10.013—the two trackers bracket rigidity to 1% over $t \leq 200$ (the long run opens the gap late; Sec. VII).

The matter survives the acceleration. The field peak is flat to 0.3% over the run, the sector rms radii grow only from 4.3 to $\sim$5.2, the minimum lapse never leaves 0.991, and min $\chi = 0.9889$—nowhere near collapse ($\chi \to 0$) or dispersal. A lone star under the same numerics holds its peak to 0.5% for $t = 200$.

TABLE I. The falsifiable run matrix: twenty-seven cells. The sign structure predicts a drifting midpoint for the mixed pair only, reversal under swapping the sectors, and a d^{-2} rate; every control row must return a null *centroid drift*—including the same-sign pairs, which merge into one lump and still do not displace their centroid. ΔX at $t = 200$ unless marked. The separation scan is $d_0 = 8, 12, 16, 20$; the mass ladder is three retuned pairs; the stability scan covers $\omega = 0.75$–0.90. Full cell names and per-cell data are in the results pack (Sec. IX).

run	grid(s)	ΔX	verdict
`PM`, d_0=10	256^3	**+3.0016**	**runaway**
`PM`, d_0=10	192^3	+3.0139	converged pair
`PM`, d_0=10	128^3	+2.8815	4% low
`PM`, d_0=8–20	128^3	—	$a \propto d^{-2.03}$
`MP` mirror	128^3	−2.8815	reversed[a]
`PM`, deeper solve	128^3	+2.8815	+0.002%
`PM`, AMR on	128^3	+2.8815	+0.001%
`PM`, box doubled	256^3	+2.7606	−4.2%
`PM`, $t \leq 400$	128^3	+11.518	a steady
`PP` $(+,+)$[b]	128^3–256^3	$\leq 7.8 \times 10^{-4}$	merges; null
`MM` $(-,-)$[b]	128^3, 192^3	$\leq 5.3 \times 10^{-4}$	merges; null
`P`, `M` lone stars[c]	128^3	$\leq 1.8 \times 10^{-3}$	null
mass ladder	128^3	—	tracks mass
stability scan	128^3	—	static

[a] Drift ratio −1.000022, acceleration ratio −1.000010.
[b] Both same-sign pairs coalesce into a single lump (Sec. V E); the entry is the pair-centroid drift over the full run, *through* that merger, which is the quantity the null tests—not a statement that the configuration is static. `MM` has two 128^3 cells, the second dumping frames.
[c] Largest core displacement on any axis over $t = 200$; the phantom sits off-centre, the sharper test.

TABLE II. The separation scan at the base grid ($N = 128$, fits over $5 \leq t \leq 200$). $a\,d^2$ should return the pair mass $\bar{M} = 0.014322$ at every separation. $|P|$ is the signed total momentum at $t = 200$—it halves with each step outward, tracking the acceleration rather than sitting at a diagnostic floor.

d_0	8	10	12	16	20
a $[10^{-4}]$	2.307	1.448	0.998	0.560	0.359
$a\,d^2/\bar{M}$	1.031	1.011	1.003	1.001	1.002
$\lvert P\rvert$ $[10^{-5}]$	5.8	3.7	2.3	0.98	0.49

B. The force law: $a\,d^2$ returns the stars' mass

Bondi's law is quantitative: each star is accelerated by the *other's* active mass, so $a = \bar{M}/d^2$. Five separations test it (Fig. 6, Table II). A power law through the five accelerations gives $a \propto d^{-2.028\pm0.011}$ against -2 exact—and adding the widest point moved the exponent *toward* the exact value (four-point fit: -2.041). The normalization is the stars' own mass: $a\,d^2/\bar{M}$ runs

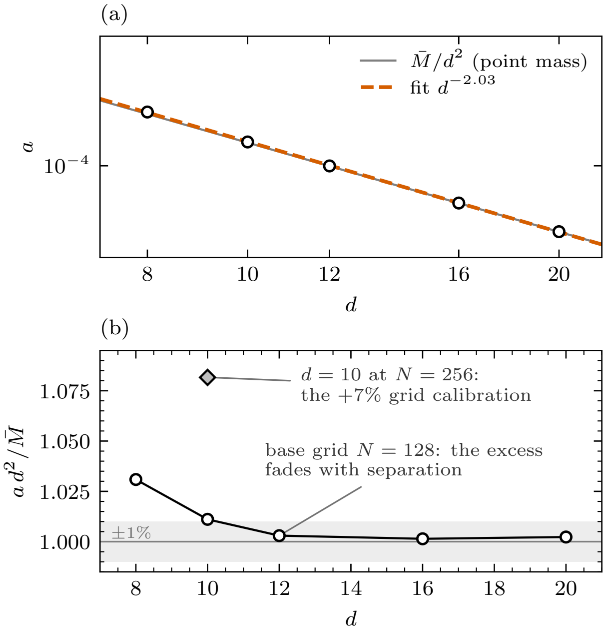

FIG. 6. The force law over $d_0 = 8$–20 at the base grid. (a) Fitted acceleration against separation (open circles): the power-law fit (orange dashed) gives $d^{-2.03\pm0.01}$, and the point-mass line $\bar{M}/d^2$ (grey) is indistinguishable from it at this scale. (b) $a\,d^2$ over the pair mass $\bar{M}$: the close-pair excess, +3.1% where the envelopes overlap most, decays monotonically and is gone by $d = 16$—a finite-size correction behaving like one. The open diamond is the converged ($N = 256$) value at $d = 10$: refining the grid raises every acceleration by ~7% (Sec. V C), a calibration common to the whole scan that cancels from the exponent and the trend.

$1.031 \to 1.011 \to 1.003 \to 1.001 \to 1.002$. The close-pair excess is physical and behaves like the finite-size correction it must be—largest exactly where the two stars' envelopes overlap most, decaying monotonically, gone by $d = 16$. The $d = 20$ cell rules out the alternative reading: had $a\,d^2$ plateaued ~1% above $\bar{M}$, the excess would have been a measurement floor. It does not, and the residual momentum halves with each step outward.

One systematic spans the whole scan: all five points share the base grid, whose acceleration at $d = 10$ sits ~7% below the converged fine-grid value (next subsection). The calibration is common mode and cancels from the exponent, from the trend in $a\,d^2$ and from every ratio in Sec. V F, but the *absolute* row of Table II carries a grid uncertainty of that size: the converged $d = 10$ acceleration, $(1.55 \pm 0.01) \times 10^{-4}$, lands 8% above $\bar{M}/d^2$, so at a gap of two stellar diameters the point-mass formula holds to ten percent rather than one. A converged separation scan is the natural sequel.

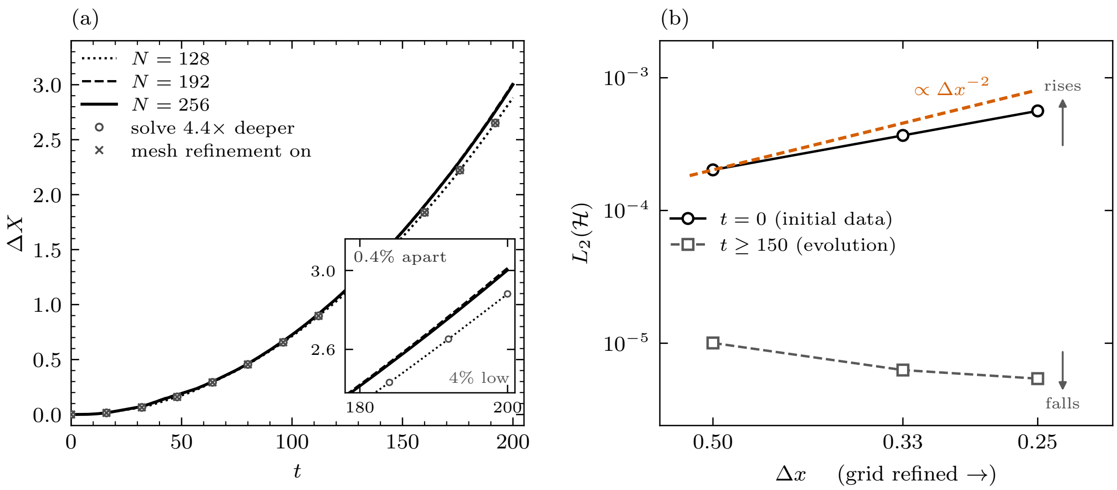

FIG. 7. (a) The resolution ladder at $d_0 = 10$: midpoint drift at $\Delta x = 0.50, 0.33, 0.25$, with two twins of the base rung overlaid—the elliptic solve driven $4.4\times$ deeper (circles) and mesh refinement enabled (crosses); both reproduce it to 2×10^{-5}. Inset: the endpoint—the two fine grids agree to 0.4% while the base rung sits 4% low. (b) The absence of a formal convergence order is physical, not a defect of the ladder: two error sources compete. The $t = 0$ violation *rises* with resolution—the solve-to-evolution handoff leaves a cell-scale residual that the second-difference constraint operator amplifies as Δx^{-2} (Appendix A)—while the evolution's own violation *falls*; two opposite grid behaviours never leave a single asymptotic power.

C. Convergence, and what cannot move the answer

Refining the grid does not erode the runaway—it settles it. The drift at $t = 200$ runs $+2.881$, $+3.014$, $+3.002$ down the ladder: the two fine rungs agree to 0.4% on drift and 0.5% on acceleration while the base rung sits 4% low, which is the opposite of what a discretization artefact does. **We quote $\Delta X(200) = 3.00 \pm 0.01$ and** $a = (1.55 \pm 0.01) \times 10^{-4}$, the fine-pair spread as the error bar. No formal convergence order accompanies this, for a measured reason (Fig. 7b): the $t = 0$ Hamiltonian norm *rises* with resolution ($2.0 \to 3.7 \to 5.6 \times 10^{-4}$, the transfer residual of Appendix A) while the evolution error falls ($1.0 \times 10^{-5} \to 5.4 \times 10^{-6}$ late-time), so the triple is non-monotone and Richardson extrapolation would return corrections smaller than the fine-pair spread at any assumed order.

Two twins of the base rung close the remaining loopholes with runs rather than arguments. *Is the base rung solve-limited?* A cell identical except for the elliptic tolerance—driven from $8.6 \times 10^{-4}\%$ to $1.9 \times 10^{-4}\%$, a $4.4\times$ deeper residual—changes the drift by 0.0015% and the acceleration by less than a part in 10^4, while refining the grid moves it by 4%: the rung offset is a property of the evolution grid, not of the solve. *Does the uniform-grid choice shape the result?* The same cell with mesh refinement enabled reproduces drift and acceleration to 0.001%–0.002%—and the log shows level 1 was never created, because the tagger's threshold ($|\chi - 1| = 0.02$) sits four times above anything these spacetimes produce. Finally, doubling the box (eight times the volume, sponge moved out) changes the drift by -4.2%: the boundary is not driving the motion.

D. The mirror: swap the sectors, the runaway reverses exactly

A drift produced by the grid, the boundary, the solver or the diagnostics would not know which star carries the minus sign; swapping the sectors in place must reverse the physics and nothing else. **It reverses to two parts in** 10^5: the mirrored cell returns drift ratio -1.000022 and acceleration ratio -1.000010 against the headline cell. The direction of travel is set by the matter configuration, not by the grid, the boundary or the gauge.

E. The nulls, and the two interaction channels

The lone-star rows calibrate the floor: a single star, alone in the box, moves at most 1.8×10^{-3} on any axis in $t = 200$ (canonical, box centre) and 1.6×10^{-3} (phantom, released *off-centre* at $x = 37$, where no symmetry protects it)—about $2000\times$ below the signal, with no preferred direction. Its field peak breathes by 0.5%.

The same-sign pairs are the campaign's sharpest control, and their behaviour was not anticipated: **they merge—both of them, on the same clock**. Tracked frame by frame (Fig. 8a), the two canonical stars of `PP` coalesce at $t = 33.6$ and the two phantoms of `MM` at $t = 32.8$—a 2.4% difference, within the trackers' grid disparity. That timing is the measurement. Newtonian gravity *attracts* the `PP` pair and *repels* the `MM` pair—Bondi's own sign rules, Fig. 1—so if gravity set the timescale, `PP` would coalesce while `MM` flew apart. Instead both collapse identically: the driving force is blind to the sign of the mass, and its scale gives it away—Newtonian free fall from $d = 10$ at these masses takes $t \approx 207$, so the

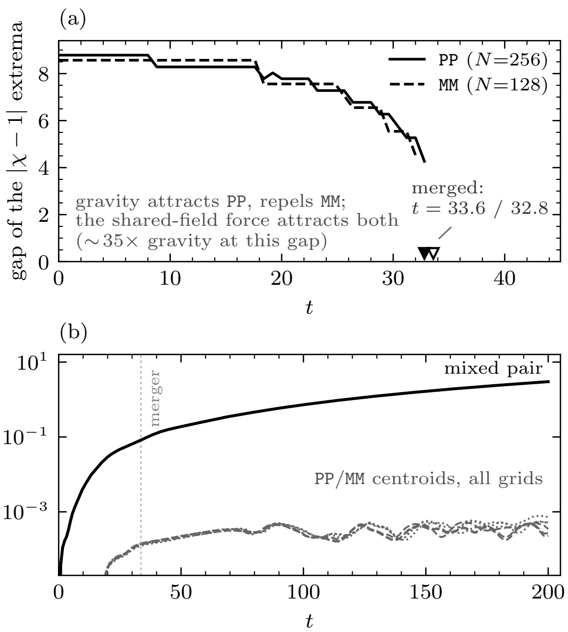

FIG. 8. The same-sign controls. (a) Gap between the two $|\chi - 1|$ extrema—wells for PP, hills for MM—tracked frame by frame: *both* pairs fall together and merge, at $t = 33.6$ and 32.8. Newtonian gravity's sign is *opposite* between these two cells, so the force that sets their common clock is blind to the sign of the mass: the shared field's overlap attraction, $\sim$35× gravity at this gap. The mixed pair has no shared field—gravity is its only channel. (b) Through plunge, merger and ringdown, the PP/MM pair centroids (all grids) stay below 8×10^{-4}—four orders under the mixed pair's drift. A configuration that cannot manufacture momentum even while merging is a stronger null than one that sits still.

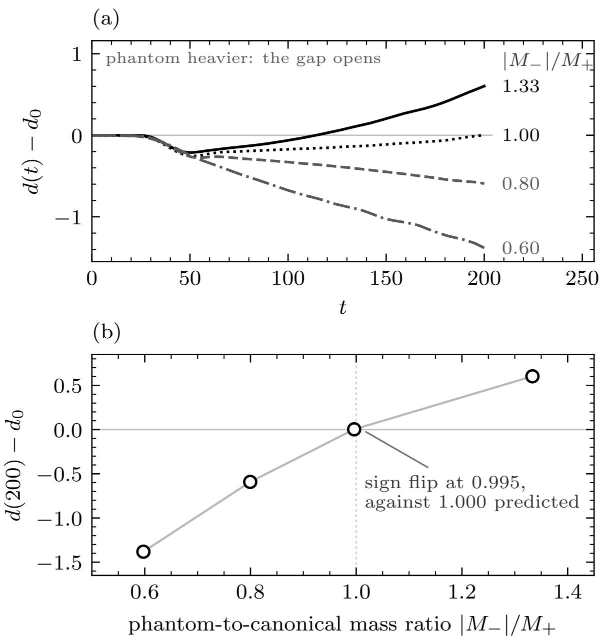

FIG. 9. The mass ladder: the matched pair and three retuned partners spanning a factor 2.2 in mass ratio. (a) Gap history: with a heavier phantom the gap *opens*, matched it holds, with a lighter phantom it *closes*, monotonically in the ratio. Every cell shares the same $t \approx 50$ gauge dip, so the gap is read late. (b) Gap change at $t = 200$ against the mass ratio, monotone through the four cells: the sign flip falls at ratio 0.995, against 1.000 predicted—the test no artefact survives, because an artefact can close a gap but cannot make the closing reverse according to which of two stars is heavier.

observed $t \approx 33$ needs a force $\sim$35–40× gravity. That is the same-field overlap attraction of Sec. II, present exactly and only where two lumps share one complex field; Bondi's $--$ repulsion is real but invisible beneath it.

This measurement is what certifies the mixed pair as gravitational. Its two stars are *different* fields with no coupling term, so the overlap channel is structurally absent, and the data confirm that absence three ways. At the same $d = 10$ where same-field pairs close in $t \approx 33$, the mixed gap holds to 1% for $t = 200$: a few-percent leak of a 35×-gravity force across sectors would have closed it visibly. The mixed force is a clean $d^{-2.03}$ power law across $d = 8$–20, where a field-overlap force dies exponentially with distance. And its strength tracks the partner's *ADM mass* (next subsection), a quantity a field force knows nothing about.

Meanwhile the same-sign cells deliver their null under the most violent history in the campaign: through plunge, merger, sevenfold ejecta growth and ringdown, their pair centroids never move more than 7.8×10^{-4} (PP) and 5.3×10^{-4} (MM) on any of six cells across three grids (Fig. 8b)—and the residual does not grow with resolution. Neither pair has a mass dipole, so neither centroid moves, *even during a merger*. The merger dynamics themselves are converged physics (ejecta growth ×7.1/7.0/6.9 across the ladder), the net inward boundary flux stays at round-off throughout, and the remnants carry the model's structural sign difference: the PP remnant is a well in χ ($\min \chi = 0.979$), the MM remnant a *hill* (χ up to 1.011)—same matter behaviour, opposite curvature. One diagnostic note: $\min \chi = 1.00000$ in MM does not mean flat geometry; a minimum only sees wells, and phantom curvature is all hills.

F. Gravity scales with the source—including its sign of inequality

Each star is accelerated by the *other's* mass, so retuning one star must change its partner's acceleration and leave its own alone. Three one-knob variants test this (Fig. 9, Table III). The retuned-partner pull ratios land within 0.9–3.1% of the partner's mass ratio across a factor 2.2, while each cell's unchanged-partner ratio—the internal control—returns 1.00–1.03. And the ladder carries a signature with no artefact analogue: **the gap's behaviour flips sign exactly where the masses cross**. Heavier phantom: the gap opens (+0.60). Matched: it holds (+0.003). Lighter phantom: it closes (−0.59,

TABLE III. The mass ladder. One star is retuned per cell; the other is untouched. "Mass factor" is the retuned star's new-to-old mass ratio, which the force law predicts to reappear as its *partner's* pull ratio—the retuned-partner force constant from a separation-corrected fit ($\ddot{x} = \pm C/d(t)^2$ per star, $t \geq 5$) over the matched cell's. The unchanged partner's ratio, predicted 1, is the internal control. Only ratios are quoted; the constants themselves absorb the common grid calibration of Sec. V B. The resulting pair ratios $|M_-|/M_+ = 0.597$, 0.800, 0.996, 1.333 are the axis of Fig. 9b.

retuned star	mass factor	pull ratio	control
Φ_-: $\omega\, 0.7603 \to 0.88$	0.600	0.618 (+3.1%)	1.028
Φ_-: $\omega\, 0.7603 \to 0.804$	0.803	0.810 (+0.9%)	1.011
— (matched)	1	1	1
Φ_+: $\omega\, 0.75 \to 0.81$[a]	0.747	0.756 (+1.1%)	1.001

[a] The phantom-heavier rung lightens the *canonical* star, leaving the harder phantom solve untouched. No rung exists at 0.40 of the matched mass: both branches floor at $|M|_{\min} \simeq 0.54$ of it (Fig. 3)—the bound is itself a result.

-1.38), monotonically. Interpolating between the two cells that bracket the flip puts it at mass ratio 0.995, against 1.000 predicted. No grid, boundary or solver distinguishes which of the two stars is heavier; gravity does.

VI. LIMITS ON GRAVITATIONAL RADIATION AND NEAR-ZONE BEHAVIOUR

A uniquely relativistic aspect of the runaway is its radiation: the exact accelerating-pair metrics are radiative [2, 3], and negative-mass binaries have been proposed as sources of anomalous signatures [39], with the leading anomalous channel for mismatched charge-to-inertia binaries being dipolar. The bicomplex model closes that channel structurally: the source of gravity is the signed stress tensor, conserved on shell (Sec. II), so the rate of change of the signed mass dipole is the conserved signed momentum—zero for this data, and measured to cancel to $\lesssim 1\%$ (Sec. V A)—and mass-dipole radiation is forbidden at leading order for exactly the reason it is forbidden in ordinary general relativity. Kinematics agrees from the other side: Ψ_4 carries spin weight -2, so its expansion begins at $\ell = 2$ whatever the dipole does. What survives is quadrupolar bremsstrahlung from the accelerating pair—secular, chirpless, with no astrophysical counterpart. At the speeds reached here that content is bounded rather than measured: every extraction shell the domain affords lies inside the near zone, so what follows is an upper limit on the $\ell = 2$ amplitude.

The doubled box measures how much of that reaches the wave zone at these speeds: **none that we can detect out to** $R = 40$ (Fig. 10). The $\ell = 2$ amplitude falls as $r^{-4.8}$ across shells spanning $R = 16$–40—$r\,\psi_4$ drops by a factor 31 where one outgoing wave would keep it flat—so every shell sits in the near zone, reading the source's

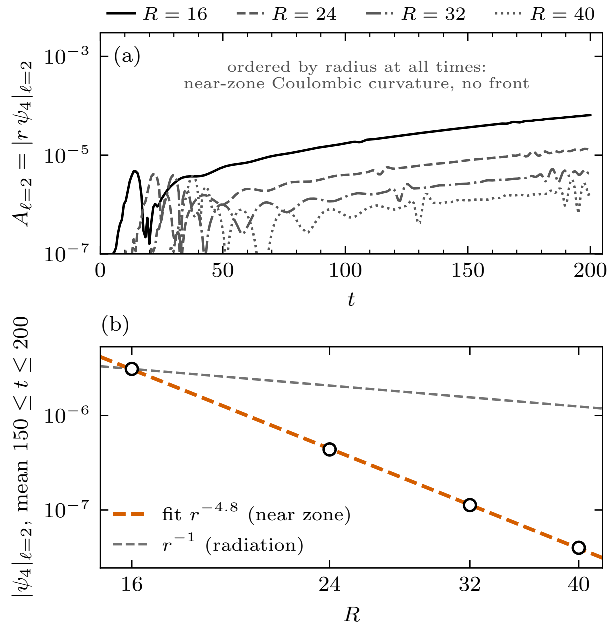

FIG. 10. The wave-zone measurement on the doubled box ($L = 128$, $\Delta x = 0.25$, sponge at 48–60). (a) Total $\ell = 2$ amplitude $A_{\ell=2}$ on the four extraction shells: ordered by radius at all times, with no propagating front. (b) The falloff, averaged over $150 \leq t \leq 200$: $|\psi_4|_{\ell=2} \propto r^{-4.8}$, against r^{-1} for radiation. Every shell out to $R = 40$ is in the near zone, so $A_{\ell=2}$ is a Coulombic near-field amplitude and not a radiated flux; no radiative tail is measurable. A null with a number, and consistent with the model's own prediction: a zero-momentum pair has a static signed mass dipole, and at $v \lesssim 0.03$ within the fit window its quadrupole flux is far below this bound.

Coulombic curvature rather than a flux. We report this as a null with a number rather than a missing measurement: it bounds the radiated content of the run, it is consistent with the structural argument above, and it defines the follow-up—evolve the pair to relativistic speed (the pair reaches only $0.03c$ inside the fit window here), where bremsstrahlung scales steeply and the energy budget of a system that begins with nearly zero total mass—and has no positive-mass theorem to floor it—becomes measurable. That budget, rather than the kinematics, is where this configuration remains open.

VII. SYSTEMATICS, WITH THEIR MEASURED SIZES

Fit convention. Fitting from $t \geq 5$ against fitting the last two thirds moves any single a by 1–3% (a late window weights the fastest part of the run); every number here uses $t \geq 5$, and the *trends*—the exponent, the ratios, the sign flip—are insensitive to the choice.

Grid calibration. The base grid underestimates the acceleration by $\sim$7% at $d_0 = 10$; the fine pair agree to 0.5% and set the quoted values. The offset is common to the (base-grid) separation and mass scans, cancelling from their differential statements; the converged absolute force law beyond $d = 10$ is unmeasured, and the converged $d = 10$ point sits 8% above $\bar{M}/d^2$—finite-size physics at a two-diameter gap, to be settled by a converged scan.

Rigidity. Constant separation is a 1%-level statement on $t \leq 200$: the two trackers bracket the gap change between -0.9% and $+0.1\%$. On the $t = 400$ window the gap opens to 11.7 while the acceleration stays steady—so "rigid pair" belongs to $t \leq 200$ and "constant rate" does not depend on it.

Coordinates and diagnostics. Drift and gap are coordinate quantities in a dynamical gauge, and the whole-domain barycentre of a weakly gravitating sector is sensitive to domain noise: that of the off-centre lone phantom wobbles by 4×10^{-2} early (domain noise entering the sector weight) while its core never leaves 1.6×10^{-3}—which is why nulls are quoted on cores and the pair on both trackers, agreeing to 1%. The measurements that do not share the coordinate choice—the mirror ratio, the signed-momentum cancellation, the same-sign merger clock, the χ hill/well structure—all corroborate the coordinate ones. One cosmetic artefact is stated for the record: a zero initial shift is inconsistent with the outer boundary condition, which emits a flat $\sim 10^{-6}$ gauge pulse inward from $t = 0$; it crosses the stars before $t = 20$ and leaves no trace in the constraint norms or the fitted acceleration.

Same-sign initial data. The conformally flat outer boundary condition of the elliptic solve is genuinely wrong for a box carrying net mass $2M$, so the same-sign solves floor near $10^{-3}\%$ rather than converging arbitrarily; this is a $t = 0$ statement only—their evolution constraints stay bounded through the merger—and it is one more reason those cells serve as momentum nulls rather than precision measurements.

VIII. DISCUSSION AND CONCLUSION

We posed Bondi's positive–negative mass pair as a full initial-value problem, and it behaved as he argued in 1957—quantitatively.

(1) The runaway is real, and it is steady. Released at rest, the mixed pair self-accelerates at a constant rate for as long as we evolve it, reaching $0.056c$ by $t = 400$ with the acceleration flat to 2%; the drift reverses under swapping the sectors to two parts in 10^5; and the separation holds to 1% over the measurement window.

(2) It is gravity. By construction the two sectors meet only in the metric; by measurement the force falls as $d^{-2.03\pm0.01}$ with $a\,d^2$ returning the stars' mass, and the pull tracks the partner's ADM mass through a factor 2.2—including the sign flip of the gap at the equal-mass point. The same-sign controls isolate the only other channel in the model, the shared-field overlap attraction, and show it to be $\sim 35\times$ gravity, short-range, and blind to the sign of the mass—present exactly where the model puts it, and absent from the mixed pair, whose gap such a leak would have closed.

(3) Displacement without momentum—the Bondi signature. The sector momenta, each growing to 5×10^{-4}, cancel in the signed sum to under 1%, and the residual converges *toward* zero with resolution while the displacement converges onto 3.00 ± 0.01. Same-sign pairs, which have no mass dipole, hold their centroids to 10^{-4}-level *even while merging*.

(4) A stable star of negative ADM mass exists as constraint-satisfying, asymptotically flat initial data and survives 400 time units of dynamical evolution, breathing at the sub-percent level (static phantom-field stars [31] and de Sitter bubbles [29] were known; a negative-mass *horizon* would be a different object entirely [12]). The phantom star is, notably, the better-behaved of the pair: shed field is pushed away rather than re-accreted, so the feedback loop that lets canonical stars deepen their own wells runs backwards.

What is not established. The continuum force-law normalization away from $d = 10$ (the scan is base-grid; Sec. VII); rigidity beyond $t = 200$, where the gap opens while the rate holds; and the radiation. On the last, the near-zone null out to $R = 40$ is consistent with the structural no-dipole argument and with quadrupole bremsstrahlung at $v \lesssim 0.03$ being unmeasurably small, but the energy budget of a radiating system whose total mass starts at 5.5×10^{-5}—0.4% of either star, with no positive-mass theorem to floor its descent—is the open question this configuration was built to reach. It needs speed: a comoving-box campaign carrying the pair to $\sim 0.3c$ is prepared, and is where the anomalous-signature programme of Ref. [39] meets a concrete source. Nothing here bears on whether such matter exists, and the contrast with Ref. [13] is environmental: immersed in the right cosmological fluid a mixed pair can bind, while our isolated, asymptotically flat pair runs away.

IX. REPRODUCIBILITY

Every number and figure derives from the packed campaign in the repository's results tree (`results/bondi-dipole-runaway/`): one directory per cell under `campaign/`, each holding the tracker streams, constraint norms, Weyl modes, solver history, and the exact launch environment. Cell names carry the configuration—`runaway_pair_d10_L64_N256_lev0` reads as the mixed pair at $d_0 = 10$ in an $L = 64$ box at 256^3, uniform grid—and `stars/` holds the dressed-star profile tables and both family scans. A single script (`make_article_figures.py`) regenerates every figure and prints every quoted number. Initial data: `GRTresna` [25]; evolution: `GRTeclyn` [23, 24, 26].

ACKNOWLEDGMENTS

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Appendix A: Constraint behaviour

Initial data. Every mixed-pair solve converges monotonically on the maximal-slicing path and exits under its gate: $8.6 \times 10^{-4}\%$ (Hamiltonian) and $8.0 \times 10^{-4}\%$ (momentum) at the base grid in 13 nonlinear iterations, tightened with the grid to $8.7 \times 10^{-5}\%/5.1 \times 10^{-5}\%$ on the finest rung (16 iterations)—so the fine rungs do not inherit the coarse rung's residual, unlike a fixed-tolerance ladder. The deep-solve twin of Sec. V C shows the evolution is insensitive to a further 4.4× of solve depth at fixed grid, because the $t = 0$ violation on the *evolution* grid is dominated by the solve-to-evolution transfer, which scales as Δx^{-2} and which no solver tolerance can touch: the deeper solve lowered it by only $\sim$1% ($2.02 \to 2.00 \times 10^{-4}$). The mechanism is the order of that handoff: the solved metric arrives as a piecewise-constant cell copy (exact once the spacings match, Sec. III B) and the matter is repainted analytically from its radial table, so the residual mismatch between them is a cell-scale quantity whose amplitude does not fall with the grid, while the operator that reads it is a second difference carrying Δx^{-2}.

Evolution. Figure 11 shows the headline ladder in absolute code units (the domain is overwhelmingly vacuum, so relative norms are dominated by near-zero denominators; the yardstick is the controls under identical numerics). Each rung relaxes from its initial-data spike within $t \lesssim 2$ and then holds nearly flat for the whole run: late-time $L_2(\mathcal{H})$ of 1.0×10^{-5}, 6.3×10^{-6}, 5.4×10^{-6} down the ladder—falling with resolution, while the $t = 0$ spike rises ($2.0, 3.7, 5.6 \times 10^{-4}$), the two-error-source structure of Fig. 7b. The `PP` control runs a factor few above the mixed cell after its merger and stays bounded to $t = 200$; its centroid null holds through that entire history.

Appendix B: Computational cost

Every evolution ran on NVIDIA H100 80 GB devices (one per run) on a node carrying four of them behind two 32-core Intel Xeon Platinum 8462Y+ processors; the elliptic solves ran CPU-side on 32 MPI ranks, $\sim$20 minutes at 256^3 and $\sim$4 hours at 512^3. `GRTeclyn` was built against AMReX with CUDA 12.1 and GCC 11.4; the commit each cell was built from is recorded in its metadata. At $\Delta t = 0.02\,\Delta x$, a $t = 200$ evolution takes 20 000/30 000/40 000 steps at $N = 128/192/256$; measured wall times are in Table IV. The twenty-seven packed cells total 81.6 GPU-hours of evolution; a two-GPU evolution path was smoke-tested and works at these sizes, but four single-GPU cells in flight is the better allocation of the node.

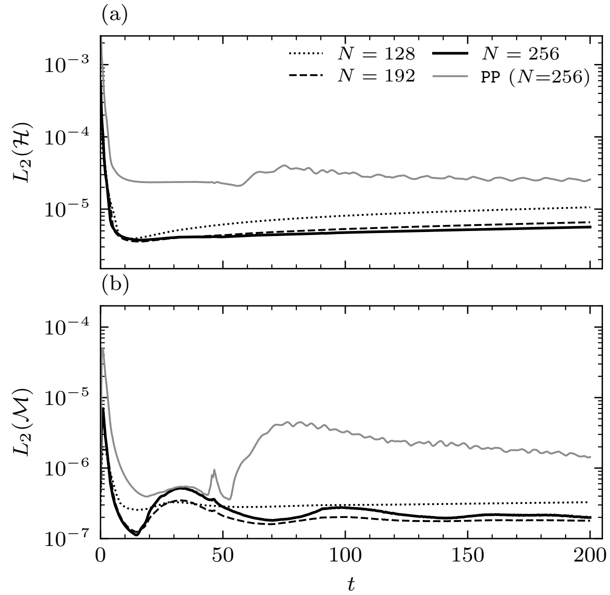

FIG. 11. Constraint norms for the headline cell's resolution ladder (line styles as in Fig. 7) and the merging `PP` control at the finest grid (gray). (a) Hamiltonian, L_2 over the domain: finer grids start worse (initial-data interpolation noise) and evolve better. (b) Momentum, L_2.

TABLE IV. Measured evolution cost per cell type (one H100 per run, $t = 200$ unless noted; "frames" adds the movie/still pipeline). Device memory is $\sim$6/20/48 GB at $N = 128/192/256$, which is what allows uniform 256^3 on one card where a refined grid would not fit.

cell type	grid	Δx	wall
pair, $L = 64$	128^3	0.50	1.1 h
pair, $L = 64$	192^3	0.33	4.8–6.0 h
pair, $L = 64$, frames	256^3	0.25	14.6 h
wave zone, $L = 128$	256^3	0.50	7.4 h
long run, $t = 400$, frames	128^3	0.50	3.8 h

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